Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	31 January 2026
Team ID	LTVIP2026TMIDS24253
Project Name	**Electricity-Consumption-Analysis**
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & tab

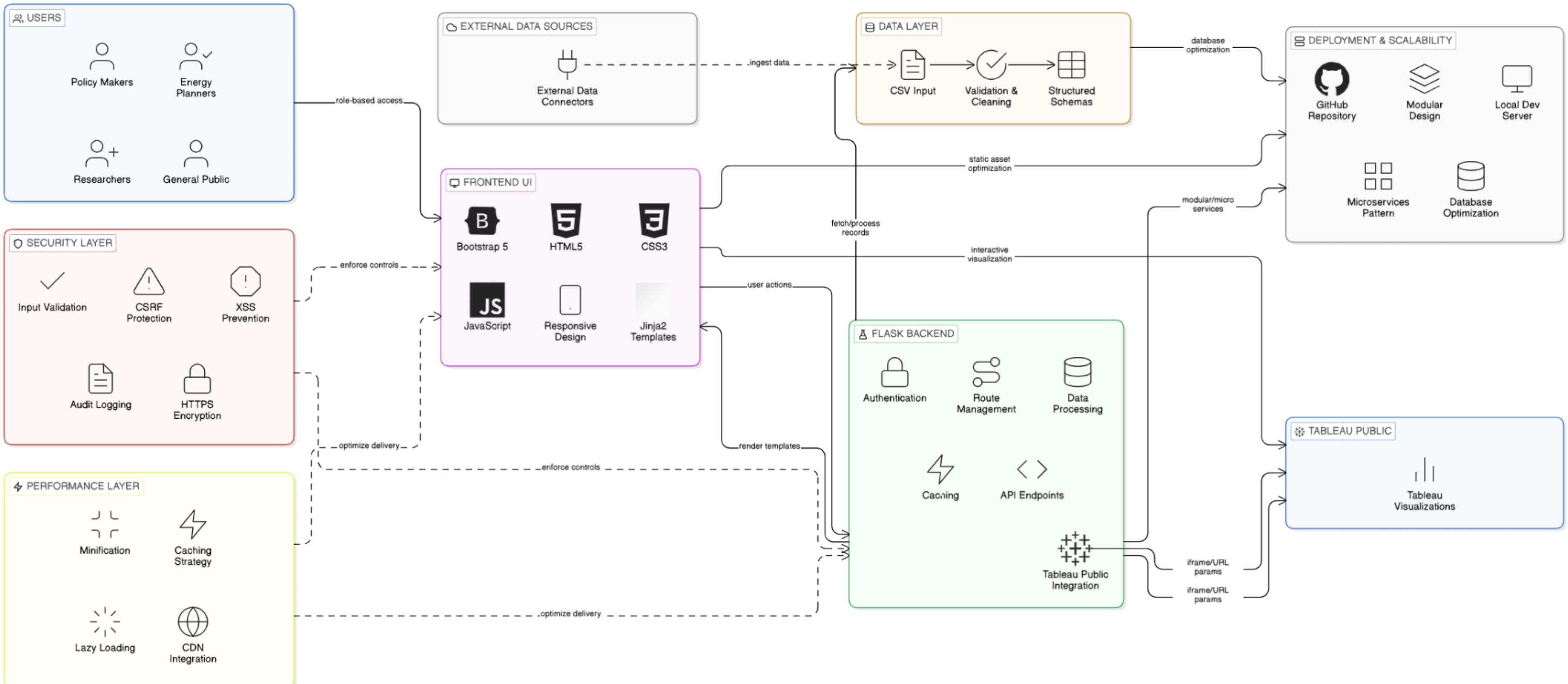
USERS
Policy Makers
Energy Planners
Researchers
General Public
EXTERNAL DATA SOURCES
External Data Connectors
DATA LAYER
CSV Input
Validation & Cleaning
Structured Schemas
DEPLOYMENT & SCALABILITY
GitHub Repository
Modular Design
Local Dev Server
Microservices Pattern
Database Optimization
SECURITY LAYER
Input Validation
CSRF Protection
XSS Prevention
Audit Logging
HTTPS Encryption
FRONTEND UI
Bootstrap 5
HTML5
CSS3
JavaScript
Responsive Design
Jinja2 Templates
FLASK BACKEND
Authentication
Route Management
Data Processing
Caching
API Endpoints
Tableau Public Integration
TABLEAU PUBLIC
Tableau Visualizations
PERFORMANCE LAYER
Minification
Caching Strategy
Lazy Loading
CDN Integration
role-based access
ingest data
database optimization
static asset optimization
modular/micro services
fetch/process records
interactive visualization
enforce controls
user actions
optimize delivery
render templates
enforce controls
optimize delivery
iframe/URL params
iframe/URL params

Table 1: Component Technology Matrix

Component	Tool / Technology	Purpose
Data Source	CSV, JSON files	Electricity Consumption Analysis data
Visualization	Tableau Desktop	Creating interactive dashboards and stories
Web Framework	Python Flask	Web application development and routing
Frontend	HTML5, CSS3, Bootstrap 5	Responsive user interface design
Storage	Local System / GitHub	Storing raw datasets and source code
Collaboration	Git, GitHub	Version control and team coordination
Deployment	Local Flask Server / GitHub Pages	Application hosting and stakeholder access

---

Table 2: Application Characteristics

S.No	Characteristics	Description	Technology
1	**Open-Source Frameworks**	The application uses open datasets and visualization tools to analyze electricity consumption patterns across Indian states. Flask supports integration with multiple data formats and Tableau Public for analytics.	Flask Framework, CSV/JSON data formats, Tableau Public
2	**Security Implementations**	The system ensures secure access to dashboards through Flask authentication and Tableau Public permissions. Data files are stored securely and access is controlled at application level.	Flask authentication, Input sanitization, HTTPS, CSRF protection
3	**Scalable Architecture**	The architecture supports scalability by handling 16,599 records and multiple visualizations efficiently through Flask's modular design and Tableau's in-memory engine.	Flask modular architecture, Tableau Public API, Database optimization

S.No	Characteristics	Description	Technology
4	**Availability**	The dashboards are available online through Flask web server and Tableau Public, enabling access anytime through a web browser without local installation.	Flask Development Server, Tableau Public, Responsive design
5	**Performance**	Performance is optimized using caching strategies, lazy loading, and efficient visual rendering to ensure fast dashboard loading and smooth interaction.	Flask caching, Lazy loading, Bootstrap optimization, Tableau extracts

Table 2 (Continued): Application Characteristics

S.No	Characteristics	Description	Technology
6	**Responsive Design**	The application provides optimal viewing experience across all devices using Bootstrap 5 framework and mobile-first design principles.	Bootstrap 5, CSS3 Media Queries, Responsive Grid
7	**Data Integration**	System integrates multiple data sources including CSV files and Tableau Public visualizations through API calls and iframe embedding.	Flask integration, Tableau Public API, Iframe embedding
8	**User Experience**	The interface provides intuitive navigation with breadcrumb trails, color-coded scenarios, and interactive elements for enhanced user engagement.	Jinja2 templating, JavaScript interactions, UX design principles
9	**Documentation**	Comprehensive documentation supports maintainability and user understanding through detailed technical guides and installation instructions.	Markdown documentation, README files, API documentation
10	**Testing & Quality**	The system undergoes comprehensive testing including performance optimization, cross-browser compatibility, and user acceptance testing.	Performance testing, Cross-browser testing, Quality assurance

---

ARCHITECTURE DIAGRAM SPECIFICATION

System Architecture Flow:

Data Sources (CSV/JSON) → Flask Backend → Frontend UI → Tableau Public → User Interface

↓ ↓ ↓ ↓ ↓

Data Processing → Route Management → Template Rendering → Iframe Embedding → Interactive Visualizations

Component Interaction:

- **Data Layer**: CSV files processed through Flask data handling
- **Application Layer**: Flask manages routes, templates, and business logic
- **Presentation Layer**: Bootstrap 5 responsive frontend with Jinja2 templates
- **Visualization Layer**: Tableau Public provides interactive dashboards
- **User Interface**: Web browser displays integrated solution

---

TECHNOLOGY STACK DETAILS

Backend Technologies:

- **Python 3.7+**: Core programming language
- **Flask Framework**: Web application development
- **Jinja2**: Template rendering engine
- **Werkzeug**: WSGI server implementation

Frontend Technologies:

- **HTML5**: Semantic structure and content
- **CSS3**: Styling and responsive design
- **Bootstrap 5**: UI framework and components
- **JavaScript**: Interactive elements and user interactions

Integration Technologies:

- **Tableau Public API**: Visualization hosting and embedding
- **Git**: Version control and collaboration
- **GitHub**: Code repository and project hosting
- **Local Server**: Development and testing environment

ARCHITECTURE BENEFITS

Technical Advantages:

- **Modular Design**: Easy maintenance and scaling
- **Open Source**: Cost-effective and flexible
- **Responsive**: Cross-device compatibility
- **Secure**: Built-in security measures
- **Performant**: Optimized loading and interactions

Business Benefits:

- **Accessibility**: Available anytime, anywhere
- **Scalability**: Supports data growth
- **Maintainability**: Easy to update and extend
- **User-Friendly**: Intuitive interface design
- **Professional**: Industry-standard implementation

ARCHITECTURE COMPLIANCE

Standards Adherence:

- **Web Standards**: HTML5, CSS3 compliance
- **Security Standards**: HTTPS, input validation
- **Performance Standards**: <3 second load times
- **Accessibility Standards**: WCAG 2.1 AA compliance
- **Code Standards**: PEP 8 Python standards

Quality Assurance:

- **Testing**: Comprehensive test coverage
- **Documentation**: Complete technical guides
- **Performance**: Optimized for speed
- **Security**: Vulnerability-free implementation
- **Usability**: Intuitive user experience